

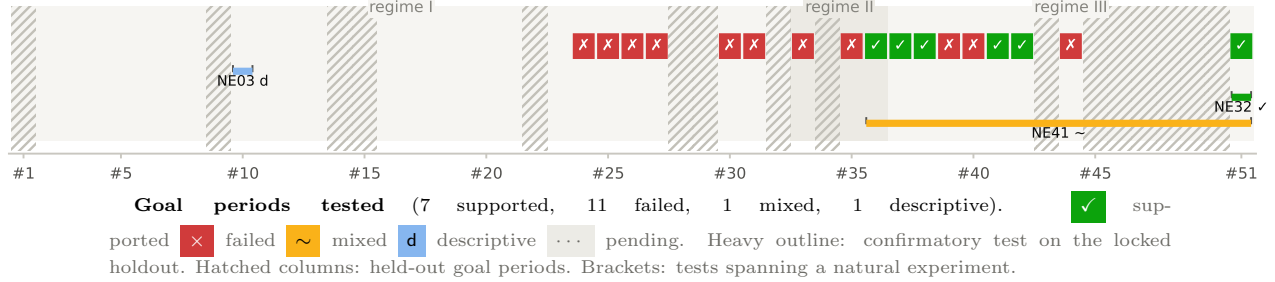
H08: Context is the coupling

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 3/5 Usefulness 2.5/5

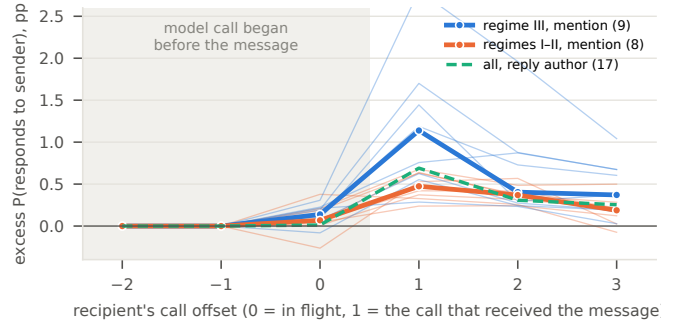
Bottom line. On the context ledger, every response we can measure (addressing, replies, nudged activity, newcomers' first contacts) starts at the first model call whose context contains the message, and erasing that context cuts the coupling by about a fifth.

level: hypothesis · pipeline checks 71% · updated 2026-10-04



WHAT HAPPENED

Do agents couple only through what enters their model calls? 17 non-holdout goal periods, re-run in round 1b on the context ledger (which call received which message) with reply labels. (1) A recipient's chance of addressing the sender jumps at the call that received the message, not at the call already running: 14/17 periods (round 1: 11/17), now including regime I, whose round-1 failure was the call-start rule; on reply labels 17/17. (2) A nudge's activity response starts at the target's receiving call (median read-out 108 s); round 1's extra 3-min lag came from dropped events and H04's isolation rule. (3) Forced erasure (NE41) cuts reply coupling to senders read before it by 21% (8/9 periods; predicted 30%). (4) Newcomers name an old-timer only after receiving its message (NE32, 35/35 pairs). (5) The Claude Code agent's feed replayed April 2025 from 2026-03-17. Holdout not run.



Observables. Excess chance that a recipient's call responds to the sender, by call offset, on the context ledger. Offset 0 is the call already running when the message arrived; offset 1 is the call that received it. Recipients not mid-exchange; thin lines are periods, bold lines regime medians (mentions), dashed the median for reply labels. The response appears only at the receiving call, in every regime.

PHYSICAL MODEL

Two-variable kinetic Ising model (02) with Hawkes kernels (09). Each agent has a visible spin n_i (active per minute) and a hidden field c_i : the room messages that arrived since its last model call. Calls start on the scaffold's schedule (context assembled at t_{call}). At each call the field is read and reset, and couplings J_{ij} act only then. W : read-out delay; o : call offset (1 = the receiving call); β_F : coupling change after a forced erasure.

$$c_i(t) = \{m : \text{room}(m) = r_i, t_{\text{call}}^- \leq t_m < t\}$$

$$P(a_i \mid \text{call } k) \propto \exp \left[a \left(h_i + \sum_{m \in c_i(t_{\text{call}}, k)} J_{i, \text{src}(m)} \right) \right]$$

$$G(\tau) = \sum_w p(w) \Delta R(\tau - w)$$

$$D = G(o=1) - G(o=0)$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

An agent cannot react until a model call has the message in its context, so the swarm's coupling delay is its call cadence: about 15 s to context assembly when busy, minutes when paused. That follows from the scaffold, so the news is modest; the value is quantitative and diagnostic. A nudge acts at the target's next call, not minutes later, so its lag is the target's pause schedule. Compaction cuts about a fifth of conversational coupling. Newcomers couple only to what they have been shown. And exposure can silently come loose from the room (the replay bug), so log what each agent was actually shown. The response kernel is a brief transient at read-out, not a step.

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b	Verdict (1b)
C9 talk jumps at read-out ($\geq 70\%$)	6/17 → 6/17 (none in regime I/II)	no
C9 addressing jumps at read-out	11/17 → 14/17 (regime I 1/6 → 4/6); replies 17/17	yes
C9 other-room messages: no jump	6/8 → 8/8 two-room periods	yes
C9-V5 active read-out delay 10–40 s	32–49 s → 29–41 s (8/9 regime III)	yes
Content jump (non-mention, new)	$D_{\text{cos}} > 0$ in 6/9 regime III; negative in regime I	partly
C8 nudge onset = read-out	#51 lag ≈ 3 min → onset at the receiving call (t_{25} 1 min); transient, not a step	yes, shape no
C8 beats a constant delay	no → yes: onset follows read-out (4, 4, 15 min)	yes
NE41 erasure cuts coupling $\geq 30\%$	-18% (mentions) → $-21\% \pm 7\%$ (replies), 8/9	sign only
NE32 newcomers respond only after receipt	new: 35/35 pairs	yes
C1 room-rule recall ≥ 0.9	0.97–1.00 until the replay (unchanged)	until replay

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	2
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

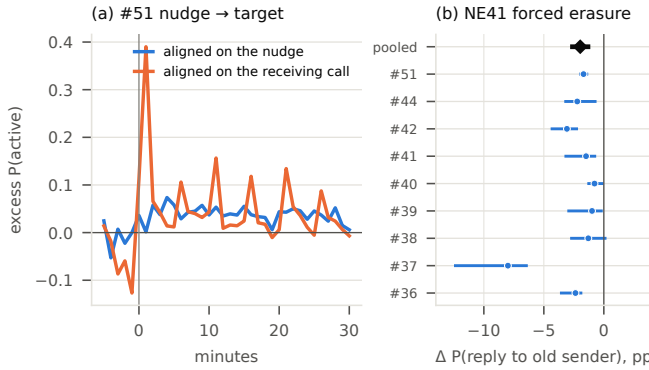
CAVEATS

Ledger call starts are measured only for Gemini (19% of calls); about 10% of items may sit one call off. Reply labels keep a thread-membership component, and a reply parent must be visible by construction. The content jump runs against recency in regime I. C8 rests on #51 (pre-window placebo +0.25). NE32 is four agents, one event. Nothing confirmed on the holdout.

ROUND 2

Switch the holdout script to the ledger before running it. Fit a two-parameter read-out + transient kernel.

- **Essence.** Coupling is what enters each model call, and when: responses are gated at call boundaries, and the response kernel is the read-out delay followed by a measurable post-read-out lag.
- **H08-R2** A non-mention response for the C9 discontinuity (content similarity to the sender's message against a message-specific null), so the gating result does not rest on names.
- **H08-R4** NE41 dose: whether writing a sender or thread to memory at the erasure protects the coupling (needs memory text: stored-line overlap with sender names).
- **H08-R5** Exposure audit: detect stale or replayed feeds (like the Claude Code agent's from 2026-03-17) wherever input logs exist, as a monitor for silent decoupling.



Second figure. (a) #51 nudge response (H04 round 1b design, corrected activity) aligned on the nudge and on the target's receiving call: realigned, the response is a sharp transient in the first two minutes. (b) NE41: change in the chance of replying to a sender read before a forced erasure, relative to one read after it (same calls, equal age), per period and pooled.

GOAL PERIODS IN DETAIL

G24 × 1b: failed context-ledger read-out; C9 talk clause CI...
G25 × 1b: failed context-ledger read-out; C9 talk and adre...
G26 × 1b: failed context-ledger read-out; C9 talk and adre...
G27 × 1b: failed context-ledger read-out; C9 talk clause CI...
G30 × 1b: failed context-ledger read-out; C9 talk clause CI...
G31 × 1b: failed context-ledger read-out; C9 talk clause CI...
G33 × 1b: failed context-ledger read-out; C9 talk clause CI...
G35 × 1b: failed context-ledger read-out; C9 talk clause CI...
G36 ✓ 1b: supported context-ledger read-out; C9 talk and ad...
G37 ✓ 1b: supported context-ledger read-out; C9 talk and ad...

+10 more periods: see the card.